UNIT I

Flexible Power Transmission & Design of Bearings

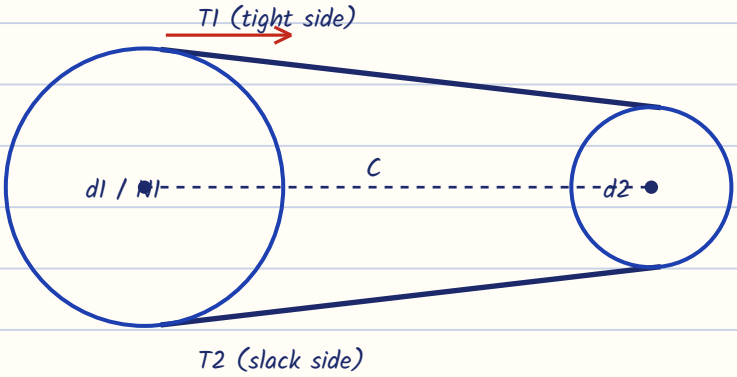
flat & V belts | pulleys | chain drives | wire ropes | lubrication | hydrodynamic theory | Sommerfeld number | McKee's equation | rolling bearings

I. Flexible power transmission – overview

Flexible drives (belts, ropes, chains) transmit power between shafts that are **far apart** where gears would be bulky. They are cheap, absorb shock and vibration, and are simple to maintain. In automobiles: **V-belt / serpentine belt** (alternator, water pump, AC compressor), **timing belt / chain** (camshaft drive), **wire ropes** (window regulators, hoists, brake cables).

Drive	Principle	Features / use
Flat belt	Friction on flat pulley	Long centre distance, high speed (up to 50 m/s), quiet; slip occurs
V-belt	Wedging in 34–40° groove	Compact, high ratio (up to 7:1), multiple belts; used in vehicles, motors
Timing (toothed) belt	Positive meshing	No slip, synchronous: cam drive, printers
Roller chain	Positive, rollers on sprocket teeth	No slip, high load, low speed, compact; bicycles, motor-cycles, timing
Wire rope	Tension member on sheave / drum	Hoists, cranes, elevators, cables over long distance

II. Belt drives – basic relations



Driver = larger pulley

Tight side is the **lower** side, slack side on top (for long drives) so wrap angle increases.

Open belt: same direction

Crossed: opposite direction

Velocity ratio, slip and creep

> **Velocity ratio** VR = N2/N1 = d1/d2 (no slip); with slip s%: N2/N1 = (d1/d2) (1 - s/100). Elastic creep adds about 0.5-1% loss.

> **Belt speed** v = π d1 N1 / 60 (m/s), usually 15-25 m/s for flat leather belts, 20-25 m/s for V-belts (max 30).

> **Creep**: relative motion because the belt stretches more on the tight side; **slip**: motion because of insufficient friction (overload).

Length of belt and angle of contact

	Open belt	**Crossed belt**
Length L	$L = \pi (r1 + r2) + 2C + (r1 - r2)^2 / C$	$L = \pi (r1 + r2) + 2C + (r1 + r2)^2 / C$
Angle of lap on small pulley θ	$\theta = 180° - 2 \sin^{-1}((r2 - r1)/C)$	$\theta = 180° + 2 \sin^{-1}((r1 + r2)/C)$
Angle on both pulleys	different (small pulley governs)	same for both pulleys

Centre distance $C \geq (d1 + d2)/2 + d1$ (roughly), and $C \geq 1.5\ d2$ for good wrap; the belt sags by about 1-2% of C.

Ratio of tensions

$$\text{Flat belt: } T1 / T2 = e^{(\mu \theta)} \qquad \text{V-belt: } T1 / T2 = e^{(\mu \theta / \sin \beta)}$$

> T1, T2 = tight & slack side tensions (N); μ = coefficient of friction; θ = angle of lap (radian); β = semi-groove angle (about 17-19° so 2β = 34-38°). The wedge makes the effective friction coefficient

$\mu/\sin\beta$ = about 3 μ - V-belt transmits much more power.

> **Power transmitted** $P = (T1 - T2)\ v$ (W); torque $T = (T1 - T2)\ d/2$.

> **Centrifugal tension** $Tc = m\ v^2$ (m = mass per metre = ρA); it lowers the useful tension: $T1 = T_{max} - Tc$, so effective ratio $(T1 - Tc)/(T2 - Tc) = e^{(\mu\theta)}$.

> **Maximum tension** $T_{max} = \sigma\ b\ t$ (allowable stress × width × thickness) ; joint efficiency 0.7-0.9 for leather (cemented 90%, laced 60-70%).

> **Initial tension** $T0 = (T1 + T2 + 2\ Tc)/2$ (= $(T1 + T2)/2$ when Tc is ignored).

> **Condition for maximum power**: $T_{max} = 3\ Tc \rightarrow v = \sqrt{(T_{max}/3m)}$.

Example 1 – flat belt power

> Belt 100 mm × 6 mm, σ_{allow} = 2 MPa, ρ = 1000 kg/m³, μ = 0.3, θ = 160° (2.79 rad), v = 15 m/s. Find power.

> $T_{max} = \sigma\ b\ t = 2 \times 100 \times 6 = 1200$ N ; $m = \rho A = 1000 \times 0.1 \times 0.006 = 0.6$ kg/m ; $Tc = m\ v^2 = 0.6 \times 225 = 135$ N.

> $T1 = 1200$ N ; $(T1 - Tc)/(T2 - Tc) = e^{(0.3 \times 2.79)} = 2.31 \rightarrow T2 = 135 + (1200 - 135)/2.31 = 596$ N.

> $P = (T1 - T2)\ v = (1200 - 596) \times 15 = 9060$ W = **9.06 kW**.

III. Design of flat belts and pulleys

Belt materials

Material	Properties / use
Leather	Oak-tanned; high friction (μ 0.3-0.4), long life, costly; $\sigma \approx 2.5$ N/mm², density ≈ 950 kg/m³.
Rubberised fabric / cotton	Cheaper, good in damp; $\sigma \approx 1.5$-2.5 N/mm²; $\mu \approx 0.3$.
Balata / synthetic (nylon core)	Strong, thin, high speed, weather proof; modern flat belts with polyamide core and chrome leather.

Design procedure (power, speeds, centre distance given)

1. Design power = P × service factor Ks (1.1 to 1.8 for light to heavy shock, long hours).
2. Select belt speed v (pulley dia from speeds); find θ on smaller pulley and μ (corrected for speed if needed).

3. Find T1 and T2 from $P = (T1 - T2)\,v$ and $T1/T2 = e^{(\mu\theta)}$ (include Tc).
4. Allowable tension per unit width: choose thickness t (ply) and σ -> width $b = T1/(\sigma\, t\, \eta_{joint})$. Round to standard width.
5. Check pulley dia ≥ minimum ($d_{min} \approx 30$-35 t for leather, 50 t for fabric) so bending stress $\sigma b = E\, t/d$ is small.
6. Length of belt by formula; initial tension; check bearing loads = T1 + T2 (vector sum).

Flat pulley design

Part	Design rule
Material	Cast iron (most), steel, wood, aluminium for light. Speed < 30 m/s (CI rim).
Rim width B	$B = 1.1\,b + 10$ mm (about 1.25 b); thickness $t_r = d/200 + 3$ mm (single belt).
Crown	Slightly convex face (crown $h \approx B/100 - B/50$) so belt remains centred.
Arms	Elliptical section, $a2 = 0.5\,a1$ (minor/major); no. of arms n = 4 (d ≤ 200 mm), 6 (d 200-1500 mm). Arms designed as cantilever beams.
Arm design	Torque T is shared by n/2 arms acting as cantilevers of length $R = d/2$: $M = T / (n/2)$ ... $M = (T1 - T2)(d/2) / (n/2)$. Section modulus $Z = \pi\, a1^2\, a2/32 = M/\sigma b$ with $a2 = 0.5\,a1$ and $\sigma b = 15$ MPa (CI); so $a1 = [\,32\,M / (\pi\,(0.5)\,\sigma b)\,]^{(1/3)}$.
Hub	Bore = shaft dia ds; hub outer dia ≈ 2 ds; hub length ≈ (1.0 to 1.5) ds (or 0.5-0.75 B); standard key as per shaft size.
Balancing	Statically / dynamically balanced for v > 10 m/s.

> Rim stress due to rotation: $\sigma = \rho\, v^2$ (same as hoop stress) < 30 MPa (cast iron)

IV. V-belts and V-pulleys

V-belts are endless, trapezoidal belts made of rubber with fabric / polyester cords and wrapped cover. Used in **multiple belts** on one drive. Standard sections per IS 2494 / SAE: **Z, A, B, C, D, E**; automotive uses **AV 10, AV 13** narrow V and **serpentine (poly-V)** belts.

Section	Top width (mm)	Thickness (mm)	Min pulley dia (mm)	Power range (kW)
Z	10	6	50-70	0.2 - 1.5
A	13	8	75-125	0.7 - 3.5
B	17	11	125-200	2 - 15

Section	Top width (mm)	Thickness (mm)	Min pulley dia (mm)	Power range (kW)
C	22	14	200-300	7 - 70
D	32	19	355-500	20 - 150
E	38	23	500+	30 - 200

> Included angle of belt = 40°, groove angle = 34°-38° (smaller for small pulleys) so belt wedges and does not bottom.

> Advantages over flat belt: compact, high velocity ratio (≤ 7), less slip, short centre distance, drive in any direction, quiet.

> Limitations: not for very long distance, cannot use crossed drive, belts must be matched sets, more bending loss.

Design steps for V-belt drive (rated-power method)

1. Design power = P × Ks (service factor).
2. Select the **section** from the chart (design power vs speed of the faster shaft).
3. Choose the pitch dia of small pulley ≥ d_min; d2 = i d1 (1 - s). Belt speed v = π d1 N1/60 ≤ 25 m/s.
4. Centre distance: 0.7 (d1 + d2) ≤ C ≤ 2 (d1 + d2); compute pitch length L, choose the nearest standard length, then correct C.
5. Angle of lap θ = 180° - 57.3 (d2 - d1)/C -> correction factor Fθ (0.85 for 150°, 1.0 for 180°).
6. Rated power per belt (table) × Fθ × FL (length factor) = corrected power per belt.
7. **Number of belts** N = design power / corrected power per belt -> round up (max 8-12 belts).
8. Calculate pulley groove dimensions and check shaft loads and initial tension.

Example 2 - V-belt, number of belts

> P = 7.5 kW, motor 1440 rpm, driven 480 rpm, service factor 1.3. Design power = 9.75 kW. Section B selected; d1 = 140 mm -> d2 = 140 × 3 = 420 mm (no slip).

> v = π × 0.14 × 1440/60 = 10.5 m/s. Assume rated power per belt at this speed = 2.6 kW; Fθ = 0.94, FL = 1.0 -> corrected 2.44 kW.

> N = 9.75/2.44 = 4.0 -> **4 belts, section B.**

V-pulley details

> Cast iron or pressed steel; groove angle 34°/36°/38° chosen with pulley dia; face width = (n - 1) e + 2f for n belts (e = groove pitch, f = edge distance).

> Pulleys must be **aligned**, belts of same make/set, tension checked after 24 hrs of running.

V. Design of chain drives

A **roller chain** has inner plates, outer plates, pins, bushes and **rollers**; it wraps on sprockets (toothed wheels). Features: positive drive (no slip), efficiency 96-98%, high load capacity, long life, works in oil/dust; noisy at high speed, needs lubrication.

ISO chain no.	Pitch p (mm)	Roller dia (mm)	Min. breaking load (kN)
06B	9.525	6.35	8.9
08B	12.7	8.51	17.8
10B	15.875	10.16	22.2
12B	19.05	12.07	28.9
16B	25.4	15.88	60
20B	31.75	19.05	95
24B	38.1	25.4	160

Basic relations

Quantity	Formula
Pitch circle diameter	$D = p / \sin(180°/z)$ (z = number of teeth)
Velocity ratio	$i = N1/N2 = z2/z1$ ($z1 \geq 17$, preferably 19-25 odd; $z2 \leq 120$; $i \leq 7$)
Chain speed	$v = z1\ p\ N1 / 60000$ (m/s); v max about 15 m/s for roller chain
Chain length in pitches	$Lp = 2\ a/p + (z1 + z2)/2 + ((z2 - z1)/2\pi)^2\ (p/a)$ (round to even number)
Centre distance	$a = (30 \text{ to } 50)\ p$, max 80 p; $a_{min} = (D1 + D2)/2 + (30\text{-}50\ mm)$
Power	$P = (T1)\ v$ where tension in chain $F = P/v$; $P_{rated} \times K$ = design power
Load / safety factor	$n = Q / (Ks\ F_t + F_c + F_s) \geq 7\text{-}20$ depending on speed

Design procedure (selection method)

1. Assume z1 (19 to 27), $z2 = i\ z1$ -> find actual ratio.
2. Design power = $P \times Ks \times K_z / K_{multi}$ (Ks 1.0-1.5 service factor; multi-strand factor 1, 1.7, 2.5 for simplex, duplex, triplex).

3. Choose chain no. from rating chart (power vs sprocket rpm) and find pitch p.
4. Choose centre distance a = 30-50 p; calculate Lp (even number of links), correct a.
5. Check chain speed, tension, breaking-load factor of safety, sprocket tooth bearing stress.
6. Select lubrication method: (i) manual drip - v < 4 m/s, (ii) oil bath/splash v 4-10 m/s, (iii) forced oil jet v > 10 m/s.

> **Chordal action (polygon effect)**: chain moves like on a polygon, so speed varies slightly; reduces with larger teeth numbers; causes noise & impact -> keep z1 ≥ 17.

> Slack side tensioner / idler sprocket; chain guard; check wear elongation: replace at 3% extension.

VI. Wire ropes

A **wire rope** is made of many steel wires twisted into **strands** that are laid around a **core**. Designation **6 × 19**: 6 strands, each with 19 wires (flexible, used for cranes, lifts); 6 × 7 stiff (haulage), 6 × 37 very flexible.

Feature	Details
Wire material	Plough steel (1770-1960 MPa), improved plough steel, galvanised for corrosion.
Core	Fibre core (hemp/ sisal, flexible, holds lubricant) or **IWRC** (independent wire rope core, stronger, for heat/crushing).
Lay	**Ordinary (regular) lay**: wires opposite to strand lay; no kinking, general. **Lang's lay**: same direction; better wear but can untwist.
Rope geometry (6 × 19)	Rope dia d; wire dia dw = 0.063 d; metallic area A = 0.4 d^2 (mm^2); weight w ≈ 0.035 d^2 N/m; Ult. breaking load Fu ≈ 0.5 d^2 kN (typical).

Stresses in a wire rope

Stress	Formula
Direct (tension)	$\sigma d = (W + w\,L + Wa) / A = W_{total} / A$ (W = load, wL = rope weight, Wa = acceleration load)
Acceleration load	$Wa = (W + w\,L)\,a / g$
Bending stress	$\sigma b = E_r \cdot dw / D$ ($E_r = 0.8 \times 2.1 \times 10^5$ MPa; D = sheave / drum dia) -> larger D, lower stress
Total (max)	$\sigma_{max} = \sigma d + \sigma b$; **FOS** = Fu / ($W_{total}$ + bending-equivalent load)
Equivalent bending load	$Wb = \sigma b\,A$

Stress	Formula
Sheave / drum diameter	D/d ≥ 18 (absolute minimum), 30-45 for cranes with 6 × 19 rope; D ≥ 400 dw for 6 × 19 (check by bending stress)

> Factor of safety (typical): 5-6 cranes, 8-12 passenger lifts, 6-8 mine hoists, 3.5-5 for slings.

> **Selection steps**: find total load -> assume FOS -> pick rope dia d from Fu ≥ FOS × W_total -> choose D/d -> check bending, wear, fatigue (number of bends) and bearing pressure on sheave p = 2 W_t/(d D) ≤ allowable (5-9 MPa cast iron).

> Care: lubricate, avoid kinks, discard when 10% of wires broken in one lay length or 10% diameter loss.

VII. Lubrication and bearing design – basics

Bearing supports a rotating shaft and carries load with little friction. **Sliding (journal / plain) bearings** work on a fluid film; **rolling (ball/roller) bearings** on rolling contact. Automobile crank, cam and connecting-rod bearings are **hydrodynamic thin-shell sliding bearings**.

Lubrication regime	Film	Remark
Hydrodynamic (thick film)	Complete fluid film generated by motion (wedge)	Friction f = 0.001-0.005; needs speed; examples: crank journals
Hydrostatic	Film by external pump pressure	Works at zero speed, high load, machine tool spindles
Boundary	Surface asperities touch, thin adsorbed layer	Starting/stopping, f = 0.05-0.2
Mixed / thin film	Part fluid, part boundary	Low speed, high load
Elastohydrodynamic (EHL)	Elastic deformation + film in rolling/gears	Rolling bearings, gears, cams

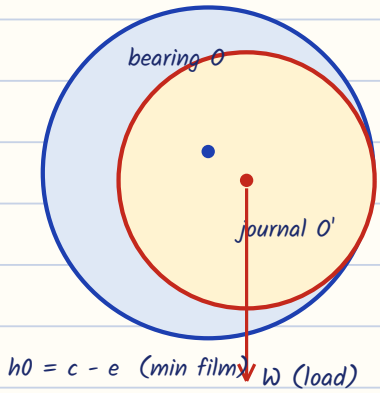
Viscosity

> Newton's law: shear stress $\tau = \mu (du/dy)$. **Absolute viscosity** μ (Pa·s = N s/m²); 1 Pa·s = 1000 cP = 10 poise; kinematic $\nu = \mu/\rho$ (mm²/s = cSt).

> Viscosity falls rapidly with temperature; **viscosity index (VI)** measures this change. SAE grades: SAE 5W-30, 10W-40, 15W-40 (engine); SAE 80W-90 (gear oil).

> **Petroff's equation** (lightly loaded, concentric journal): friction torque $T = 4\pi^2 \mu N r^3 l / c$; coefficient of friction $f = 2\pi^2 (\mu N/p)(r/c)$.

VIII. Hydrodynamic lubrication theory



c = radial clearance = R - r

e = eccentricity

ε = e / c (eccentricity ratio)

h0 = c (1 - ε)

At rest the journal touches the bearing; with speed, oil is dragged into the wedge and it **floats**.

Reynolds (1886) showed that a rotating journal drags oil into the converging wedge, creating a pressure that lifts it. **Osborne Reynolds' equation** for a one-dimensional (infinitely long) bearing is:

$$d/dx\ (\ h^3 / \mu \cdot dp/dx\ ) = 6\ U \cdot dh/dx$$

(U = surface velocity, h = film thickness)

> **Assumptions**: Newtonian incompressible lubricant, laminar flow, constant viscosity, no slip at the surfaces, pressure constant across film, negligible inertia & curvature, no side leakage (for 1-D).

> Pressure is max just after minimum film; full 3-D theory (finite bearing, l/d = 1) solved numerically by **Raimondi & Boyd charts** (variables: ε, h0/c, f r/c, Q/(r c N l), Ps etc. versus S).

> **Minimum film thickness** must exceed the sum of surface roughness: h0 ≥ 0.002 mm (≈ 5-10 times roughness) for safety.

> **Sommerfeld** solved Reynolds' equation exactly for the 'infinite bearing' (1904).

Design parameters (dimensionless)

Quantity	Formula / typical value
Unit bearing pressure	$p = W / (l\ d)$ (N/mm²); for automobile crank 7-14 MPa (up to 25 in main bearing)
Clearance ratio	$c / r = 0.001$ (to 0.002); $c = (d_{bearing} - d_{journal})/2$
Slenderness ratio	l / d = 0.5 to 2 (common 1 for crank: 0.5-0.75, machine 1-2)
Sommerfeld number	$S = (\mu N / p)(r / c)^2$ (N in rev/s, μ in Pa·s, p in Pa) - dimensionless
Eccentricity ratio	ε = e / c = 1 - h0 / c
Friction variable	f r/c = φ (S, l/d) from chart (Raimondi & Boyd)

Quantity	Formula / typical value
Heat generated	$Hg = f W v = f W (\pi d N)$ (W); $v = \pi d N$ (m/s, N rev/s)
Heat dissipated	$Hd = C A (t_b - t_a)$; $A = l d$ (projected area); C = 140-420 W/m²°C (still air), 490-1400 (good air flow)

Interpretation of Sommerfeld number

- **High S** (high speed, low load, high viscosity): thick film, small eccentricity ratio; **low S** (low speed, heavy load): film thin, $\varepsilon \to 1$, metal contact possible.
- Practical designs use S about 0.04 to 0.15 (stable side of the minimum-friction point); S below this is unsafe.
- Raimondi-Boyd chart gives h0/c, f r/c, flow Q, max pressure ratio for each S and l/d.

IX. Design of a sliding contact bearing using Sommerfeld number

1. Given: load W, journal dia d (from shaft design), speed N, operating temperature & lubricant (SAE grade) -> viscosity μ at operating temp.
2. Choose l/d (1 to 1.5), bearing material, allowable pressure p; $l = (l/d)\, d$; $p = W/(ld)$ must be ≤ allowable (check Table).
3. Choose clearance ratio c/r = 0.001 (or c = 0.001 d)(typical 0.0005 - 0.002).
4. Compute Sommerfeld number $S = (\mu N/p)(r/c)^2$; read from chart: h0/c, f r/c and flow for l/d.
5. Find $h0 = (h0/c)\, c$ and verify $h0 \geq (h0)min = 0.005 + 0.00004\, d$ (mm) ... i.e. ≥ about 0.02 mm.
6. Friction $f = (f\, r/c) / (r/c)$; heat generated $Hg = f W v$.
7. Heat dissipated $Hd = C A (tb - ta)$ with $A = l d$; check Hd ≥ Hg; if not, add cooling or oil circulation (or change l/d, c).
8. Final check: temperature rise of oil $\Delta t = Hg / (\rho Cp Q)$ with $\rho Cp \approx 1.76 \times 10^6$ J/m³K.

Bearing material	Max pressure (N/mm²)	Max temp (°C)	Use
Tin / lead babbitt	6 - 10	130	Crank, main bearings (thin lining on steel)
Copper-lead	14 - 20	150	Heavy diesel engines
Phosphor bronze	14	200	Gudgeon pins, gear boxes

Bearing material	Max pressure (N/mm²)	Max temp (°C)	Use
Aluminium-tin	28	150	Modern engine shells
Cast iron / carbon-graphite	3.5 – 7	up to 400	Slow speed, dry

X. Design using McKee's equation

McKee & McKee (1932) experimentally found that the coefficient of friction in a journal bearing varies with the **bearing characteristic number Z N / p** as shown by a curve with a **minimum friction point** – the point of transition between thick film and mixed lubrication.

$$\mu = K\,(Z\,N / p)(d / c) + k \quad \text{(McKee's equation)}$$

- Z = absolute viscosity (kg/m·s = Pa·s), N = speed (rpm), p = bearing pressure (N/mm²), d = journal dia, c = diametral clearance (same unit as d).
- K = 33 × 10^{-8} (constant) ; k = factor for end leakage: k = 0.002 for l/d from 0.75 to 2.8 (k depends on l/d; 0 for very long bearings).
- **Bearing modulus** K_b = value of (Z N/p) at the **minimum friction point**. For safe design operate at Z N/p ≥ 3 K_b – i.e. well in the thick film region – so that small changes of load or temperature do not cause film failure (metal contact).
- Heat generated Hg = μ W v (W); dissipated Hd = C A (t_b – t_a). With McKee, equate Hg = Hd to find operating temperature; check.
- K is read from the measured μ vs ZN/p curve (data-book); the margin of 3 ensures the bearing stays on the safe side of the minimum-friction point.

Example 3 – McKee check

> Journal d = 75 mm, l = 75 mm, W = 12 kN, N = 1500 rpm, Z = 0.02 kg/m·s (20 cP), diametral clearance c = 0.075 mm (d/c = 1000). $p = W/(l\,d)$ = 12000/(75 × 75) = 2.13 N/mm².

> $Z\,N / p$ = 0.02 × 1500 / 2.13 = 14.1. $\mu = 33 \times 10^{-8} \times 14.1 \times 1000 + 0.002 = 0.0047 + 0.002 =$ **0.0067**.

> $v = \pi\, d\, N/60$ = 5.89 m/s ; $Hg = \mu\, W\, v$ = 0.0067 × 12000 × 5.89 = **474 W**. $A = l\, d$ = 5.625 × 10^{-3} m²; with C = 1400 W/m²°C (ventilated): $Hd = 7.9\,(tb - ta)$ -> $tb - ta$ = 60 °C. Acceptable only with good oil circulation.

XI. Selection of rolling contact bearings

Type	Load capacity	Features / use
Deep-groove ball	Radial + moderate axial both ways	Most common; high speed, quiet; gearbox, wheels, motors
Angular contact ball	Radial + heavy axial (one way)	Used in pairs; machine tool spindles, pinion shafts
Self-aligning ball / spherical roller	Radial; accepts misalignment	Long shafts, heavy loads
Cylindrical roller	High radial, no/limited axial	Gearbox, rolls, heavy radial
Tapered roller	Radial + axial (one way)	Automobile wheel hubs, differential, pinion; mounted in pairs
Needle roller	High radial in small space	Universal joints, rockers, gearbox
Thrust ball / roller	Pure axial	Clutch release, vertical shafts

Life and capacity

Term	Meaning / formula
Basic rating life L10	Life (million revs) that 90% of identical bearings will reach or exceed (10% failure)
Basic dynamic load rating C	Constant radial load for which L10 = 1 million revolutions (from catalogue)
Life equation	L10 = (C/P)^k million revs; k = 3 (ball), 10/3 (roller)

Term	Meaning / formula
Life in hours	$L10h = 10^6 \, L10 / (60 \, n) = (10^6 / 60 \, n)(C/P)^k$
Equivalent dynamic load	$P = X \, V \, Fr + Y \, Fa$ (V = 1 inner ring rotating, 1.2 outer ring rotating)
Shock factor	$P' = Ks \, P$; Ks = 1.0 (steady), 1.2-1.5 (light shock), 1.5-3 (heavy shock)
Reliability factor	$L_R = a1 \, L10$; a1 = 1 (90%), 0.62 (95%), 0.21 (99%) (adjusted life)
Static load C0	Max load for which permanent deformation < 0.0001 d_roller; used for slow / oscillating
Varying loads	Equivalent load $P = [\, \Sigma (P_i^k \, n_i \, t_i) / \Sigma (n_i \, t_i) \,]^{(1/k)}$

Application	Required life L10h (hrs)
Automobile wheel / gearbox bearings	1500 - 5000 (about 3×10^5 km)
Light machinery, household	4000 - 8000
General industrial 8 hr/day	12,000 - 20,000
Continuous 24 hr (pumps, fans)	40,000 - 60,000

Selection procedure

1. Find radial & axial loads at the bearing (from gear forces), speed n, bore (shaft dia d).
2. Choose type (Table above) and required life L10h (application).
3. Compute equivalent load $P = X \, V \, Fr + Y \, Fa$ (X, Y from catalogue vs Fa/Fr and e).
4. Required dynamic capacity $C = P \, (60 \, n \, L10h / 10^6)^{(1/k)}$ (k = 3 ball, 10/3 roller).
5. Select a bearing (series, bore) from catalogue with $C_{catalogue} \geq C$ required; check static capacity and limiting speed.
6. Select fits (inner ring on shaft: k5/m5; outer in housing: H7), lubrication (grease/oil), sealing, preload, mounting arrangement (one fixed, one floating).

Example 4

> Deep-groove ball bearing, Fr = 4 kN, Fa = 0 -> P = 4 kN; n = 1200 rpm; L10h = 8000 hr.

> $L10 = 60 \times 1200 \times 8000/10^6 = 576$ million rev ; $C = P \times (L10)^{(1/3)} = 4 \times (576)^{(1/3)} = 4 \times 8.32 =$ **33.3 kN**.

> From the catalogue pick a deep-groove bearing with $C \geq 33.3$ kN, e.g. **6210 (C ≈ 35 kN)** (check actual catalogue value).

XII. Quick revision & important questions

Formulas to remember

> T1/T2 = e^(μθ) (flat); e^(μθ/sinβ) (V); P = (T1 - T2) v; Tc = m v²; max power at v = √(T_max/3m).

> Chain: D = p/sin(180°/z); v = z p N/60000; Lp = 2a/p + (z1 + z2)/2 + ((z2 - z1)/2π)² p/a.

> Rope: σd = W_total/A; σb = E_r dw/D.

> Sommerfeld S = (μN/p)(r/c)²; McKee μ = 33 × 10⁻⁸ (ZN/p)(d/c) + k; design at ZN/p ≥ 3K.

> Rolling bearing: L10 = (C/P)^k; P = X V Fr + Y Fa.

1. Derive the ratio of belt tensions for a flat belt and a V-belt. (5 M)
2. Design a flat belt drive to transmit given power; also design the pulley. (10 M)
3. Explain the design of a V-belt drive. Selection of number of belts. (10 M)
4. Design a roller chain drive for given power and speeds. (10 M)
5. A wire rope is used in a hoist; find rope size and stresses. (10 M)
6. Explain hydrodynamic lubrication theory with Reynolds equation. (10 M)
7. Design a journal bearing using Sommerfeld number / McKee's equation. (10 M)
8. How do you select a rolling contact bearing? Explain L10 life. (10 M)